AWS INFRASTRUCTURE RESILIENCE

HIGH AVAILABILITY ON AWS

Auto-Scaling & Multi-AZ Recovery

CORE ARCHITECTURAL CHALLENGE

Keeping a virtual server running — and what happens to its data — when an entire availability zone goes down.

FOUNDATIONS

THE BASICS: WHAT AUTO-SCALING DOES

Multiple Subnets

Auto-scaling can start EC2 instances across several subnets, not just a single subnet.

Automatic Failover

If an entire availability zone has an outage, a new virtual server can be launched automatically into another subnet in another zone.

Continuous Monitoring

Auto-scaling watches the health of running servers and reacts immediately when something goes wrong in the pool.

ARCHITECTURE CONFIGURATION

SETTING IT UP

COMPONENT 01

Launch Configuration: Blueprint

Everything needed to launch a single virtual server:

- Image (AMI) to start the new server from
- Storage, key pairs, security groups & user data

COMPONENT 02

Instance Type & Sizing

The computational capacity of the virtual server:

- vCPU count and RAM allocation based on workload
- EBS-optimized bandwidth and networking performance

COMPONENT 03

Auto-Scaling Group: Fleet Manager

Tells EC2 how many servers to run and where:

- Min / max / desired number of virtual servers
- Subnets across multiple AZs to launch servers in

COMPONENT 04

Health Checks & Lifecycle

Automatic failure detection and instance recycling:

- EC2 status checks and ELB application health checks
- Grace period management during boot time

WORKFLOW & EXECUTION

PUTTING IT TOGETHER

1. Configuration

- Launch Configuration defines AMI, size, storage & security
- Auto-scaling group assigns multi-AZ subnets & limits

2. Health Monitoring

- Continuous health ping to instances every 30s
- Zone outage immediately marks instances unhealthy

3. Automatic Failover

- New instance booted in healthy alternative AZ
- Desired capacity restored without manual intervention

COMPARATIVE ANALYSIS

TWO DIFFERENT RECOVERY MECHANISMS

SAME AZ ONLY

CloudWatch Alarm Recovery

- Launches an identical copy of the failed virtual server
- Recovers ONLY in the same Availability Zone

Why only the same zone?

- The private IP address and the EBS volume are hard-bound to a single subnet and zone.
- FAILS if an entire AZ suffers a catastrophic outage.

MULTI-AZ RESILIENT

Auto-Scaling Failover

- Recovers the failed virtual server into a completely different Availability Zone.
- Provides TRUE protection against data center outages.

Production Example: Jenkins CI Server

- If a team relies on Jenkins to test, build & deploy, auto-scaling migrates execution into another zone instead of remaining offline during an outage.

CRITICAL ARCHITECTURAL PITFALL

PITFALL: NETWORK-ATTACHED STORAGE



SOLUTIONS & STRATEGIES

THREE WAYS TO RECOVER YOUR DATA

OPTION 01

Multi-AZ Managed Service

- Use Amazon RDS Multi-AZ, DynamoDB, or S3.
- AWS handles synchronous data replication and DNS failover automatically.
- Zero maintenance burden.

OPTION 02 • RECOMMENDED

EBS Snapshots

- Snapshot EBS volume periodically.
- Snapshots are stored in S3, making them available across all Availability Zones in the region.
- Restore new EBS volume from the snapshot in Zone B on boot.

OPTION 03

Distributed Storage

- GlusterFS, DRBD, Ceph, or MongoDB replica sets.
- Replicates data directly across AZ instances.
- Very powerful, but introduces high operational complexity.

CASE STUDY IN PRODUCTION

RECOVERY STRATEGY FOR JENKINS

1

The Constraint

Jenkins stores all build configs and historical logs on the local disk.

If the disk is lost, pipeline configs are lost.

2

Plain EBS Limit

An EBS volume exists only in one single AZ.

When that AZ dies, the EBS volume cannot be accessed in Zone B.

3

Skip Distributed

Distributed clusters (GlusterFS) work, but require 3x instances and constant triage.

Unjustified complexity.

4

The Solution

EBS Snapshots!

Hourly snapshots to S3.

When ASG recovers in Zone B, create volume from latest snapshot.

Minimal data loss!

SUMMARY

KEY TAKEAWAYS

- Auto-scaling needs two pieces: a Launch Configuration (what to launch) and an Auto-Scaling Group (how many, where, healt
- CloudWatch alarm recovery only repairs instances in the same AZ; Auto-Scaling provides true cross-AZ failover.
- EBS volumes are bound to a single availability zone — they NEVER automatically follow a server into another zone.
- Bridge the storage gap with Multi-AZ managed services, EBS snapshots in S3, or distributed file systems.
- **For disk-based workloads like Jenkins, EBS snapshots hit the sweet spot: real resilience without distributed overhead.**